Cheatsheet AZ-104 — 11 · App Service: plans, scaling, slots, domains, TLS, backup and networking

1. App Service plan: tiers and what each one unlocks

Tier	Max. instances	Slots	What it adds	Signal
Free (F1)	1 (shared)	0	60 min CPU/day, 1 GB; no custom domain or SLA	"test at no cost"
Shared (D1, Windows only)	1 (shared)	0	Custom domain; no custom TLS; CPU quota	Rare
Basic (B1-B3)	**3**	0	Dedicated compute, **domain + TLS**, **Always On**, **manual** scaling, **VNet integration**, **private endpoint**, hybrid connections, automatic and custom backups (production slot only)	"custom domain, minimal cost"
Standard (S1-S3)	**10**	**5**	**Autoscale**, **deployment slots**, Traffic Manager	"staging slot", "autoscale"
Premium v2 / v3 / v4 (P0v3-P5mv3, Pv4...)	**30**	**20**	More power, **zone redundancy**, automatic scaling, more slots	"99.99 %", "zone redundant", "faster"
Isolated v2 (I1v2-I6v2)	**100**	20	**App Service Environment v3**: isolated network inside your VNet, maximum scale	"fully isolated", "no shared infrastructure"

- The plan sets the **OS (Windows / Linux), region, tier, size and instances**; all apps in the plan share the instances and scale together (optional per-app scaling). To **isolate** a heavy app or scale it separately → **new plan**. An app can only be moved to a plan in the **same region, RG and OS**.
- **Scale up** = change tier (more CPU / RAM / features; the whole plan). **Scale out** = more instances (manual or autoscale). App Service autoscale = Azure Monitor autoscale (cheatsheet 09): rules on **CPU %, memory %, HTTP queue length, Data In/Out** or another resource's metric, schedule-based profiles, cooldown. **Automatic scaling** (Premium v2+): the platform scales per app with "maximum burst" and "always ready instances", no rules.
- Default domain: `https://<app>.azurewebsites.net` (new apps carry a unique suffix `<app>-<hash>.<region>-01.azurewebsites.net`). Globally unique name.

2. Deployment slots

Concept	Detail
What it is	A live app with its own hostname (`<app>-<slot>.azurewebsites.net`) in the **same plan**; Standard **5**, Premium / Isolated **20**; no extra cost
Swap	Exchanges **content and configuration** with production **without downtime** (warm-up first); **rollback** = swap again. Usually with **production as the target** (any two slots can be swapped)
Swap with preview	Two phases: applies the production configuration to staging for validation, then completes (or cancels)
Auto swap	Every deployment to the slot is swapped automatically (CI/CD); **not on Linux** or containers
Traffic routing	% of production traffic sent to a slot (canary, A/B); `x-ms-routing-name` cookie; `?x-ms-routing-name=staging` for manual opt-in
Swapped	Framework and version, 32/64-bit, WebSockets, **app settings and connection strings** (unless sticky), handler mappings, public certificates, WebJobs content, hybrid connections, service endpoints, path mappings
Not swapped (slot-specific)	**Custom domains**, private certificates and TLS bindings, **scale settings**, publishing endpoints, WebJobs schedulers, **IP restrictions**, Always On, diagnostic logs, CORS, HTTPS only / TLS version, **managed identities**, **VNet integration**, settings marked "Deployment slot setting"

- "Connection string must point to the test DB in staging and prod DB in production" → mark the setting as a **Deployment slot setting** (sticky). "Validate with the production settings before going live" → swap with preview. "Send 10 % of users to the new version" → traffic %.

3. Domains and certificates

Task	How	Detail
Custom domain	Custom domains → Add: prove ownership with **TXT** `asuid.<domain>` = Custom Domain Verification ID, and **CNAME** `www` → `<app>.azurewebsites.net` (subdomain) or **A** record to the inbound IP + TXT (root)	Basic tier or higher for HTTPS; added to **each slot** (not swapped); also for Traffic Manager (CNAME to the profile)
App Service domain	Buy a domain from the portal (managed in Azure DNS)	"purchase a domain in Azure"
Certificates	**App Service managed certificate** (free, automatic renewal, Basic+, no wildcard, not exportable) · **App Service certificate** (purchased, stored in Key Vault, wildcard supported) · **Import from Key Vault** (grant access to the "Microsoft Azure App Service" principal) · **Upload PFX** · public certificate (.cer) so the app trusts others	Import from Key Vault = centralized renewal
TLS binding	**SNI SSL** (free, several certificates per IP; the usual choice) or **IP-based SSL** (dedicated IP; cost; for clients without SNI)	"dedicated IP address for TLS" → IP-based
Protocols	**HTTPS Only**, minimum version **TLS 1.2** (1.3 available), client certificates (mTLS)	"redirect HTTP to HTTPS" → HTTPS Only

4. Backup and restore

Type	Tier	What it includes	Details
Automatic backups	Basic, Standard, Premium, Isolated	Content and configuration, up to **30 GB**	**Hourly**, **30-day** retention, nothing to configure and no storage account; cannot be downloaded; restore over the app, to another app or to a slot
Custom backups	Basic (production only), Standard, Premium, Isolated	Content + configuration up to **10 GB**; a **linked database** can no longer be added to new configurations (existing ones keep working until March 31, 2028)	Needs a **storage account** (SAS) and container; **on demand or scheduled** (min. every 2 h, 12 per day); retention 0-30 days or indefinite; **partial backups** with `_backup.filter`; backup to firewalled storage via VNet integration; for the database, use its **native backup**

- Restoring **stops** the app during the process → restore to a **slot** first and swap. "Backup must include the database" → the database's native backup (Azure SQL, flexible server), not the custom backup. "Backup fails: content exceeds 10 GB" → exclude folders with `_backup.filter` or move files to Storage.

5. Networking

Direction	Feature	What it does	Tier
Inbound	**Access restrictions**	List of allow / deny rules by **IP**, **service tag** or **subnet (Microsoft.Web service endpoint)** with priority; once the first one is created, everything else is denied; also for the **SCM** site	All
Inbound	**Private endpoint**	Private IP in your VNet; combined with "public network access disabled" the app is no longer reachable from the internet; DNS `privatelink.azurewebsites.net`	Basic+
Inbound	**App Service Environment v3**	The whole infrastructure inside your VNet (ILB or external)	Isolated v2
Outbound	**VNet integration** (regional)	The app egresses into your VNet through a subnet **delegated** to `Microsoft.Web/serverFarms` (/26 recommended; up to 2 integrations per plan, shared by its apps); reaches VMs, databases with private endpoint, on-prem via VPN / ER; **Route All** to force all traffic; NAT Gateway for a fixed outbound IP	Basic+
Outbound	**Hybrid Connections**	TCP to an on-prem host:port via Relay, no VPN, with the Hybrid Connection Manager installed there	Basic+
Outbound	**Outbound IPs**	Listed in Properties (they change when the tier changes); for whitelists in external firewalls (or NAT Gateway for a fixed one)	All

- "App must reach a SQL server on-premises without a VPN" → Hybrid Connections. "App must call a database with private endpoint" → VNet integration. "Only the corporate IP range can reach the app" → access restrictions. "App must not be reachable from the internet at all" → private endpoint + public access disabled (or ASE).

6. Configuration, deployment and monitoring

Area	What shows up
App settings / connection strings	Environment variables encrypted at rest; sticky per slot; Key Vault references `@Microsoft.KeyVault(SecretUri=...)` with managed identity
General settings	Stack and version, 32/64-bit, **Always On** (keeps the app from unloading when idle; Basic+; required for continuous WebJobs), **ARR affinity** (sticky sessions; disable in stateless apps), HTTP 2.0, WebSockets, FTP state (FTPS only / disabled), basic auth for publishing
Path mappings	Virtual directories; **mount Azure Files / Blob** as a path
Authentication (Easy Auth)	Managed login with Entra ID, Microsoft, Google, Facebook, Apple, OpenID Connect without code; "require authentication"
Managed identity	System or user-assigned for Key Vault, Storage, SQL, ACR
Deployment Center	GitHub Actions, Azure Repos, Bitbucket, **Local Git**, FTPS, **ZIP deploy** (`az webapp deploy --src-path app.zip`), `az webapp up`, container from ACR / Docker Hub (single or Compose), **Run From Package** (`WEBSITE_RUN_FROM_PACKAGE=1`)
Logs	App Service logs: application logging (Filesystem 12 h / Blob), web server logging, detailed errors, failed request tracing, deployment logs; **Log stream**; **Kudu** (`<app>.scm.azurewebsites.net`: console, processes, files); diagnostic settings to Log Analytics; **Application Insights**
Health check	Path the platform probes; unhealthy instances are taken out of rotation (≥ 2 instances) and replaced
Diagnose and solve problems	Automatic detectors for CPU, memory, 5xx, slowness

7. Commands that show up

Task	Azure CLI	PowerShell (Az.Websites)
Plan and app	`az appservice plan create -n plan1 -g RG1 --sku S1 --is-linux` · `az webapp create -n app1 -g RG1 --plan plan1 --runtime "DOTNETCORE:8.0"` (`az webapp list-runtimes`)	`New-AzAppServicePlan -Name plan1 -ResourceGroupName RG1 -Location westeurope -Tier Standard -NumberofWorkers 2 -WorkerSize Small` · `New-AzWebApp -Name app1 -ResourceGroupName RG1 -AppServicePlan plan1`
Scale	`az appservice plan update -n plan1 -g RG1 --sku P1V3 --number-of-workers 3` · autoscale: `az monitor autoscale create --resource plan1 --resource-type Microsoft.Web/serverfarms --min-count 2 --max-count 10 --count 2` + `az monitor autoscale rule create --condition "CpuPercentage > 70 avg 10m" --scale out 1`	`Set-AzAppServicePlan -Name plan1 -ResourceGroupName RG1 -Tier PremiumV3 -WorkerSize Medium -NumberofWorkers 3`
Settings	`az webapp config appsettings set -n app1 -g RG1 --settings KEY=value --slot-settings STICKY=1` · `az webapp config connection-string set --connection-string-type SQLAzure --settings Db="..."` · `az webapp config set --always-on true --min-tls-version 1.2 --ftps-state Disabled` · `az webapp update --https-only true`	`Set-AzWebApp -Name app1 -ResourceGroupName RG1 -AppSettings @{KEY="value"} -HttpsOnly $true -AlwaysOn $true` · `Set-AzWebAppSlotConfigName -AppSettingNames STICKY`

Task	Azure CLI	PowerShell (Az.Websites)
Slots	`az webapp deployment slot create -n app1 -g RG1 --slot staging --configuration-source app1` · `az webapp deployment slot swap -n app1 -g RG1 --slot staging --target-slot production [--action preview]` · `az webapp traffic-routing set --distribution staging=10`	`New-AzWebAppSlot -Name app1 -ResourceGroupName RG1 -Slot staging` · `Switch-AzWebAppSlot -SourceSlotName staging -DestinationSlotName production -Name app1 -ResourceGroupName RG1` · `Add-AzWebAppTrafficRouting`
Domain and TLS	`az webapp config hostname add --webapp-name app1 -g RG1 --hostname www.contoso.com` · `az webapp config ssl upload --certificate-file c.pfx --certificate-password ...` · `az webapp config ssl bind --certificate-thumbprint <t> --ssl-type SNI` · `az webapp config ssl create` (managed) · `az webapp config ssl import --key-vault kv1 --key-vault-certificate-name cert1`	`Set-AzWebApp -HostNames @("www.contoso.com","app1.azurewebsites.net")` · `New-AzWebAppSSLBinding -Name www.contoso.com -ResourceGroupName RG1 -Thumbprint <t> -SslState SniEnabled -WebAppName app1`
Backup	`az webapp config backup create -n app1 -g RG1 --container-url "<sas>" --backup-name b1` · `az webapp config backup update --frequency 1d --retain-one true --retention 30` · `az webapp config backup restore` · automatic: `az webapp config snapshot list / restore`	`New-AzWebAppBackup -Name app1 -ResourceGroupName RG1 -StorageAccountUrl "<sas>"` · `Edit-AzWebAppBackupConfiguration -FrequencyInterval 1 -FrequencyUnit Day -RetentionPeriodInDays 30` · `Restore-AzWebAppBackup`
Networking	`az webapp vnet-integration add -n app1 -g RG1 --vnet vnet1 --subnet sub-apps` · `az webapp config access-restriction add -n app1 -g RG1 --rule-name office --action Allow --ip-address 203.0.113.0/24 --priority 100` · `az network private-endpoint create ... --group-id sites` · `az webapp hybrid-connection add`	`Add-AzWebAppAccessRestrictionRule -ResourceGroupName RG1 -WebAppName app1 -Name office -Priority 100 -Action Allow -IpAddress 203.0.113.0/24`
Deploy and logs	`az webapp deploy -n app1 -g RG1 --src-path app.zip --type zip` · `az webapp up` · `az webapp deployment source config --repo-url ... --branch main` · `az webapp log config --application-logging filesystem --level information` · `az webapp log tail` · `az webapp restart`	`Publish-AzWebApp -ResourceGroupName RG1 -Name app1 -ArchivePath app.zip` · `Restart-AzWebApp`

8. Decision tree

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What does the wording ask for?
├─ Custom domain and HTTPS at minimal cost ............................. Basic
├─ Staging slot / autoscale / scheduled backups ........................ Standard
├─ Zone redundancy, 20 slots, more power ............................... Premium v3 / v4
├─ Full network isolation .............................................. Isolated v2 (ASE v3)
├─ One app needs more CPU than the rest ................................ New plan (scale up or isolate)
├─ More capacity by schedule / load .................................... Scale out with autoscale (Standard+)
├─ Deploy without downtime and be able to roll back .................... Slot + swap (rollback = swap again)
├─ Different setting in staging and production ......................... Deployment slot setting (sticky)
├─ Test the new version with some of the users ......................... Traffic routing %
├─ Free, auto-renewed certificate ...................................... App Service managed certificate
├─ Centrally managed certificate ....................................... Import from Key Vault
├─ Legacy clients without SNI .......................................... IP-based SSL
├─ Downloadable or scheduled backup to a storage account ............... Custom backup (Basic+)
├─ Office IPs only ..................................................... Access restrictions
├─ Nothing from the internet ........................................... Private endpoint + public access disabled
├─ Reach VNet resources / on-prem via VPN .............................. VNet integration (delegated subnet)
├─ Reach an on-prem server without VPN ................................. Hybrid Connections
├─ Secrets without putting them in settings ............................ Key Vault references + managed identity
└─ The app "sleeps" and is slow to respond ............................. Always On (Basic+)
```

9. Signals → answer

Wording	Answer
"deployment slots" + "Basic plan"	Upgrade to Standard (5 slots)
"need 8 slots"	Premium (20)
"scale to 15 instances"	Premium (Standard tops out at 10)
"autoscale based on CPU"	Standard or higher + autoscale rule on the plan
"move the app to a plan in another region / OS"	Not possible: create a new app (same region, RG and OS to move)
"after the swap the app points to the test database"	The connection string was not sticky: mark it "Deployment slot setting"
"custom domain missing after swap"	Expected: domains are not swapped; add it to the slot
"validate the production configuration before completing the swap"	Swap with preview
"domain verification fails"	Missing the `asuid.<domain>` TXT or the CNAME / A record
"root domain (contoso.com) to the app"	A record to the IP + asuid TXT (or alias in Azure DNS)
"free certificate for www.contoso.com"	App Service managed certificate (no wildcard)
"wildcard certificate"	App Service certificate (purchased) or your own PFX
"restore the app without downtime"	Restore to a slot and swap
"backup every 6 hours to a storage account"	Scheduled custom backup (Basic+, every 2 h minimum); the database with its native backup
"automatic backups retained 30 days"	Included in Basic+ (hourly)
"allow traffic only from the Application Gateway"	Access restriction by service tag / subnet + service endpoint

Wording	Answer
"app cannot resolve the private endpoint of the database"	Missing VNet integration (and private DNS)
"app needs a static outbound IP"	NAT Gateway on the integration subnet
"WebJob stops running"	Enable Always On
"user sessions lost when scaling out"	ARR affinity (or better: external state)

10. Mistakes to avoid

1. Looking for slots or autoscale in Basic, or more than 10 instances in Standard.
2. Scaling an app "separately" without moving it out of the plan: scale out affects every app in the plan.
3. Forgetting to mark environment settings as sticky before the swap, or expecting the domain and certificate to travel with it.
4. Confusing VNet integration (outbound) with private endpoint (inbound) or with App Service Environment.
5. Using Hybrid Connections when a VPN and VNet integration already exist (or the reverse: asking for a VPN for a single TCP port).
6. Trying to move the app to a plan in another region, RG or OS.
7. Counting on the linked database backup: it can no longer be configured and goes away in 2028.
8. Leaving basic FTP and HTTP allowed when the ask is "secure": FTPS only, HTTPS only, TLS 1.2.